

Saif Ur Rehman

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MSc Data Science & AI candidate with research and hands-on experience across large language models, agentic AI and retrieval-augmented generation (RAG). Comfortable across the full research loop — reviewing academic and industry literature, designing experiments to evaluate AI models and tools, and turning findings into clear written summaries and recommendations. Leads two research programmes (protein foundation models and 4D reconstruction) and co-authored a study on reward misalignment in multi-agent LLM evaluation, alongside a commercial applied-AI internship. Expert in Python and PyTorch; builds LLM and RAG prototypes with LangChain and Qdrant, and brings an earlier career as a senior cloud and systems engineer. Open to fractional / part-time engagement (1–2 days per week) alongside the MSc.

EDUCATION

University of Liverpool — MSc Data Science and Artificial Intelligence Sep 2025 – 2026
Expected: Distinction Liverpool, UK

- Dissertation: *Beyond AlphaFold: Filling the Blind Spots in Protein Structure Prediction*. Coursework in Machine Learning and Bio-inspired Optimisation, Computational Intelligence, and Research Methods in Computer Science.

NUST, Pakistan — BEng Electrical & Telecommunication Engineering 2016 – 2021

RESEARCH EXPERIENCE

Separating Solvers and Evaluators Is Not Enough: Peer-Reward Misalignment in Multi-Agent LLMs 2026
Co-author • submitted, NeurIPS 2026 workshop (AIGood) Multi-agent LLMs • reward modelling

- Showed that separating solver and evaluator roles does not by itself remove incentives for reward manipulation in multi-agent LLM evaluation.
- Designed the experiments, agent prompting strategies, and evaluation pipelines over model outputs and agent trajectories, targeting systematic rather than anecdotal failure modes.

Beyond AlphaFold — Geometry-Aware Data Curation for Protein Foundation Models 2026
MSc research • PyTorch, OpenFold, MMseqs2, SLURM/multi-GPU github.com/saifrhman/702_BeyondAF

- Built *PDBClean*, a configuration-driven 22-stage curation framework over a fixed Protein Data Bank snapshot, published as versioned reproducible releases rather than a one-off dataset.
- Implemented a Backbone Rigid Invariant duplicate-detection cascade — length bucketing, SciPy cKDTree prefilter, a from-scratch compressed cover tree (Elkin & Kurlin 2023), exact L_∞ classification — and found a notation error in the source paper while reimplementing it.
- Curated ~499,770 retained chains from ~578,524 eligible by removing ~78,754 geometric near-duplicates, with full lineage from raw structure to released record.
- Retraining OpenFold end-to-end on the Barkla2 HPC cluster (SLURM, GPU): extraction, filtering, MSA generation, training, checkpointing, inference and evaluation.
- Built a browser-based inspection tool with Mol* 3D visualisation and lineage views, making curation decisions auditable rather than opaque.

Geometric Evidence Utilization in Feed-Forward 4D Reconstruction 2026 – present
First-author paper in preparation, targeting CVPR 2027 3D/4D understanding • world models

- Investigating whether feed-forward 4D reconstruction models convert stronger geometric conditioning into correct estimates of a target physical quantity; structured as Diagnose → Explain → Fix.
- Diagnosed state-of-the-art models (Any4D, 4RC): a ground-truth-pose intervention showed the conditioning pathway is functionally active while target tracking stays near zero, isolating the failure to evidence utilisation rather than availability.
- Designing KINETIC, an end-to-end architecture representing dynamic scenes as independently transforming kinematic components with a separated global prediction gauge; geometry-only stages complete, RGB integration in progress.
- Ran the programme under a strict development firewall: one transformation family and 20 locked test scenes held out from all training, tuning and model development for a single final evaluation.

INDUSTRY EXPERIENCE

Rofsix — Applied AI / Machine Learning Intern May 2026 – Present

- Developing a real-time system that detects and redacts confidential information during screen sharing, combining OCR, transformer-based NER and image processing.
- Built and evaluated inference components with ONNX, OpenVINO and PaddleOCR; profiled the pipeline and applied quantisation, selective processing, change detection and parallel execution for low-latency CPU deployment.
- Work across Python and C#/.NET codebases, debugging model integration and communicating experimental results to engineering stakeholders.

Creative Tech Solutions — Data Science Intern Jan 2024 – Apr 2024

- Trained a PyTorch classifier to route service-desk tickets into operational categories, replacing manual first-pass triage; evaluated with precision/recall and error analysis over ambiguous and under-represented classes.

RoshTech — Senior Systems Engineer

Nov 2022 – Sep 2025

- Built data-driven threat-detection and anomaly-analysis workflows over MongoDB, Graylog and the Elastic Stack, supporting production incident investigation for enterprise clients.

Earlier: Senior Cloud Engineer, Tier3tech • L2 Support Team Lead, MiGo Innovations.

SELECTED PROJECTS

Retrieval-Augmented Generation & Sample-Efficient Learning

f1-ai-copilot • active-learning-classification

LangChain • Qdrant • embeddings • prompt engineering • scikit-learn

- Grounded RAG system over FIA Formula 1 regulations (ingestion, chunking, embedding, top- k retrieval, cited generation), kept modular so retrieval and prompting vary independently.
- Closed-loop active learning scoring an unlabelled pool by predictive uncertainty: 0.986 test accuracy with 60 labelled examples against 70 for random acquisition at identical budgets.

TacticLens — Feedback-Refined Tactical Retrieval

github.com/saifrhman/tacticlens

Python • multi-agent spatiotemporal data • similarity search • human-in-the-loop ranking

- Retrieval over multi-agent football tracking data returning moments whose spatiotemporal configuration is structurally similar, rather than matching hand-built lexical tags; analyst judgement is a persisted input that measurably changes later rankings.
- Public repository is a separate architectural reference implementation; the original commercial system is subject to confidentiality and is not published.

rt-cais — Real-Time Conversational AI Inference System

github.com/saifrhman/rt-cais

Python • PyTorch • FastAPI • Redis • latency observability

- Serving architecture for early intent and sentiment prediction on partial utterances: async gateway, rolling-context session state, and an orchestrator applying confidence thresholds and fallback decisions.
- Separates orchestration, state and inference so model behaviour evolves independently of weights; instrumented for p50/p95 latency, error rate, drift and fallback frequency.

Hybrid Receipt Extraction — Noise-Robust OCR + Document AI

github.com/saifrhman/noise-robust-ocr-pipeline

EasyOCR • LayoutLMv3 • rule/model fusion • experiment tracking

- Fuses a deterministic rule parser with LayoutLMv3 token classification under configurable confidence thresholds, with weak-label filtering, structured ablation reports and checkpoint promotion.

Football Intelligence Platform

github.com/saifrhman/football-analytics-platform

BigQuery • dbt • Docker • FastAPI • CI/CD — end-to-end platform: bronze ingestion, silver normalisation, dbt gold models with tests, served through FastAPI and Streamlit, containerised with CI.

Parallel Monte Carlo on Barkla HPC

github.com/saifrhman/barkla-openmp-mpi-cuda-pi

C++ • OpenMP • MPI • CUDA • SLURM — shared-memory, distributed and GPU implementations of the same estimator with SLURM batch scripts and comparative scaling measurements.

COMPETITIVE PROGRAMMING & HACKATHONS

- **ICPC NWERC 2025** — University of Liverpool (team Baby Harp Seal); 107th across Northwestern Europe. **ICPC UK & Ireland 2025** — 50th nationally, 1st at Liverpool.
- **Liverpool Computer Society Hackathon** — 1st place, deep-learning solution built in competition. **Ultamation Hackathon** — 2nd place, constraint-satisfaction optimisation with dependency-aware scheduling.

TECHNICAL SKILLS

- **Languages:** Python, C++, C#, SQL, Bash, JavaScript/TypeScript, R
- **LLMs, GenAI & agentic systems:** large language models (ChatGPT, Claude, Gemini), prompt engineering, multi-agent LLM systems, reward modelling, retrieval-augmented generation (RAG), LangChain, vector search (Qdrant), Hugging Face Transformers, model evaluation & benchmarking
- **ML training & evaluation:** PyTorch, multi-GPU and SLURM training, checkpointing, large-scale data curation, controlled baselines, ablation studies, error analysis, active learning, Bayesian optimisation (BoTorch)
- **Vision & multimodal:** 3D/4D reconstruction, geometric deep learning, OpenCV, YOLOv8, U-Net, LayoutLMv3, OCR, nearest-neighbour search at scale
- **Systems, cloud & compute:** cloud platforms (AWS, Azure), SLURM, CUDA, MPI, OpenMP, HPC clusters, ONNX, OpenVINO, Docker, FastAPI, Git, CI/CD, Linux, PostgreSQL, MongoDB

CERTIFICATIONS

Machine Learning Specialization, Stanford • IBM Data Science Professional Certificate • Fundamentals of Accelerated Computing with Modern CUDA C++, NVIDIA • Associate Data Engineer in SQL, DataCamp